2 The geometry of linear programming

- **Introduction**
  - the feasible set in a linear programming problem is a polyhedron
  - a nonempty polyhedron has at least one corner point if and only if it does not contain a line, and if this is the case, optimal solutions can be found in corner points.
- **Polyhedra and convex sets**
  - **Polyhedron**
    - $\{\mathbf{x} \in \mathbb{R}^n | \mathbf{A}\mathbf{x} \geq \mathbf{b}\}$, $\mathbf{A} : m \times n$
    - standard form: $\{\mathbf{x} \in \mathbb{R}^n | \mathbf{A}\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$
    - infinite or finite is okay
  - **bounded**: A set $S \subset \mathbb{R}^n$ is **bounded** if there exists a constant K such that the absolute value of every component of every element of S is less than or equal to K.
  - **hyperplane**
    - $\{\mathbf{x} \in \mathbb{R}^n | \mathbf{a}'\mathbf{x} = b\}$
    - $\mathbf{a}$ is **perpendicular** to the hyperplane
  - **halfspace** $\{\mathbf{x} \in \mathbb{R}^n | \mathbf{a}'\mathbf{x} \geq b\}$
  - **Convex Sets**
    - **Definition**: A set $S \subset \mathbb{R}^n$ is **convex** if for any $\mathbf{x}, \mathbf{y} \in S$, and any $\lambda \in [0, 1]$, we have $\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} \in S$
    - geometry: a set is **convex** if **the segment** joining any two of its elements is **contained** in the set.
  - $\sum_{i=1}^{k} \lambda_i \mathbf{x}^i$ is a **convex combination**, where $\sum_{i=1}^{k} \lambda_i = 1$
  - **convex hull**: the set of all convex combinations of these vectors
- **Extreme points, vertices, and basic feasible solutions**
  - They are equivalent.
  - **Extreme points**
    - a point that cannot be expressed as a convex combination of two other elements of the polyhedron
    - **Definition(entirely geometric):** Let P be a polyhedron. A vector $\mathbf{x} \in P$ is an **extreme point** of P if we cannot find two vectors $\mathbf{y}, \mathbf{z} \in P$, both different from $\mathbf{x}$, and a scalar $\lambda \in [0, 1]$, such that $\mathbf{x} = \lambda\mathbf{y} + (1 - \lambda)\mathbf{z}$.
  - **Vertex**: **Definition(geometric):** A vector $\mathbf{x} \in P$ is a **vertex** of P if there exists some $\mathbf{c}$ such that $\mathbf{c}'\mathbf{x} < \mathbf{c}'\mathbf{y}$ for all $\mathbf{y}$ satisfying $\mathbf{y} \in P$ and $\mathbf{y} \neq \mathbf{x}$.
  - **active**: If a vector $\mathbf{x}^*$ satisfies $\mathbf{a}_i'\mathbf{x}^* = b_i$ for some i in M_1, M_2, or M_3, we say that the corresponding constraint is **active** or **binding** at $\mathbf{x}^*$.
    $$\begin{aligned} \mathbf{a}_i'\mathbf{x} &\geq b_i, & i \in M_1, \\ \mathbf{a}_i'\mathbf{x} &\leq b_i, & i \in M_2, \\ \mathbf{a}_i'\mathbf{x} &= b_i, & i \in M_3, \end{aligned}$$
  - **Theorem 2.2** Let $\mathbf{x}^*$ be an element of $\mathbb{R}^n$ and let $I = \{i | \mathbf{a}_i'\mathbf{x}^* = b_i\}$ be the set of indices of constraints that are active at $\mathbf{x}^*$. Then, the following are equivalent
    - There exist **n vectors** in the set $\{\mathbf{a}_i | i \in I\}$, which are **linearly independent**
    - The **span** of the vectors $\mathbf{a}_i, i \in I$, is **all of** $\mathbb{R}^n$, that is, every element of $\mathbb{R}^n$ can be expressed as a linear combination of the vectors $\mathbf{a}_i, i \in I$
    - The system of equations $\mathbf{a}_i'\mathbf{x} = b_i, i \in I$, has **a unique solution**
  - **Corner point** **Definition(algebraic):** a feasible solution at which there are n linearly independent active constraints.
  - **Basic solution**
    - All **equality constraints** are active
    - Out of the constraints that are active at $\mathbf{x}^*$, there are n of them that are **linearly independent.**
  - **Basic feasible solution** if $\mathbf{x}^*$ is a basic solution that satisfies all of the constraints.
  - **Theorem 2.3** Let P be a nonempty polyhedron and let $\mathbf{x}^* \in P$. Then following are equivalent
    - the proof is so meaningful!
    - $\mathbf{x}^*$ is a vertex
    - $\mathbf{x}^*$ is an extreme point
    - $\mathbf{x}^*$ is a basic feasible solution
  - **Adjacent basic solutions** $n - 1$ linearly independent constraints that are active at both of them.
- **Polyhedra in standard form**
  - **Theorem 2.4** Consider the constraints $\mathbf{A}\mathbf{x} = \mathbf{b}$ and $\mathbf{x} \geq \mathbf{0}$ and assume that the $m \times n$ matrix $\mathbf{A}$ has linearly independent rows. A vector $\mathbf{x} \in \mathbb{R}^n$ is a basic solution if and only if we have $\mathbf{A}\mathbf{x} = \mathbf{b}$, and there exist indices $B(1), ...B(m)$ such that:
    - The columns $\mathbf{A}_{B(1)}, ..., \mathbf{A}_{B(m)}$ are linearly independent;
    - If $i \neq B(1), ..., B(m)$, then $x_i = 0$
  - Procedure for constructing **basic solutions**
    - 1. Choose m linearly independent columns $\mathbf{A}_{B(1)}, ..., \mathbf{A}_{B(m)}$
    - 2. Let $x_i = 0$ for all $i \neq B(1), ..., B(m)$
    - 3. Solve the system of m equations $\mathbf{A}\mathbf{x} = \mathbf{b}$ for the unknowns $x_{B(1)}, ..., x_{B(m)}$.
    - nonnegative $\rightarrow$ basic feasible solution
  - Terminology
    - basic variables: the variables $x_{B(1)}, ..., x_{B(m)}$
    - nonbasic variables: the remaining varibles
    - basic columns: the columns $\mathbf{A}_{B(1)}, ..., \mathbf{A}_{B(m)}$ (linearly independent, form a basis of $\mathbb{R}^m$)
    - basic indices: $\{B(1), ..., B(m)\}$
    - **basis matrix**: $m \times m$ matrix $\mathbf{B}$ $\quad \mathbf{B}\mathbf{x}_B = \mathbf{b} \rightarrow \mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b}$
  - Correspondence of bases and basic solutions
    - geometry
    - different basic solutions must correspond to different bases
    - two different bases may lead to the same basic solution ($\mathbf{b} = \mathbf{0}$)
  - Adjacent basic solutions and adjacent bases standard form problems: two **bases** are **adjacent**, if they share all but one basic column
  - The full row rank assumption on A As long as the feasible set is nonempty, a linear programming problem in standard form can be reduced to an equivalent standard form problem(with the same feasible set) in which equality constraints are linearly independent.
- **Degeneracy**
  - **Definition 2.10**: A basic solution $\mathbf{x} \in \mathbb{R}^n$ is **degenerate** if **more than n** of the constraints are active at $\mathbf{x}$.
  - judgement of nondegenerate basic feasible solution
    - find the active constraints
    - judge the number of linearly independent constraints
  - Degeneracy in standard form polyhedra **Definition 2.11**:Consider the standard form polyhedron $P = \{\mathbf{x} \in \mathbb{R}^n | \mathbf{A}\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$ and let $\mathbf{x}$ be a basic solution. Let m be the number of rows of $\mathbf{A}$. The vector $\mathbf{x}$ is a **degenerate** basic solution if more than $n - m$ of the components of $\mathbf{x}$ are zero.
  - Degeneracy is not a purely geometric property if a basic feasible solution is degenerate under one particular standard form representation, then it is degenerate under every standard form representation of the same polyhedron.
- **Existence of extreme points**
  - **Definition 2.12**: A polyhedron $P \subset \mathbb{R}^n$ **contains a line** if there exists a vector $\mathbf{x} \in P$ and a nonzero vector $\mathbf{d} \in \mathbb{R}^n$ such that $\mathbf{x} + \lambda\mathbf{d} \in P$ for all scalars λ.
  - **Theorem 2.6**: Suppose that the polyhedron $P = \{\mathbf{x} \in \mathbb{R}^n | \mathbf{a}_i'\mathbf{x} \geq b_i, i = 1, ..., m\}$ is nonempty. Then, the following are equivalent:
    - The polyhedron P has at least one extreme point
    - The polyhedron P does not contain a line
    - There exist n vectors out of the family $\mathbf{a}_1, ...\mathbf{a}_m$, which are linearly independent
  - **Corollary 2.2**: Every nonempty **bounded polyhedron** and every nonempty polyhedron in standard form has at least one basic feasible solution.
- **Optimality of extreme points**
  - **Theorem 2.7** Consider the linear programming problem of minimizing $\mathbf{c}'\mathbf{x}$ over a polyhedron P. Suppose that P has at least one extreme point and that there exists an optimal solution. Then, there exists an optimal solution which is an extreme point of P.
    - interesting
  - **Theorem 2.8** Consider the linear programming problem of minimizing $\mathbf{c}'\mathbf{x}$ over a polyhedron P. Suppose that P has at least one extreme point. Then, either the optimal cost is equal to $-\infty$, or there exists an extreme point which is optimal.
    - useful proof
- **Representation of bounded polyhedra** **Theorem 2.9**: A nonempty and bounded polyhedron is the convex hull of its extreme points.
- **Projections of polyhedra: Fourier-Motzkin elimination***